



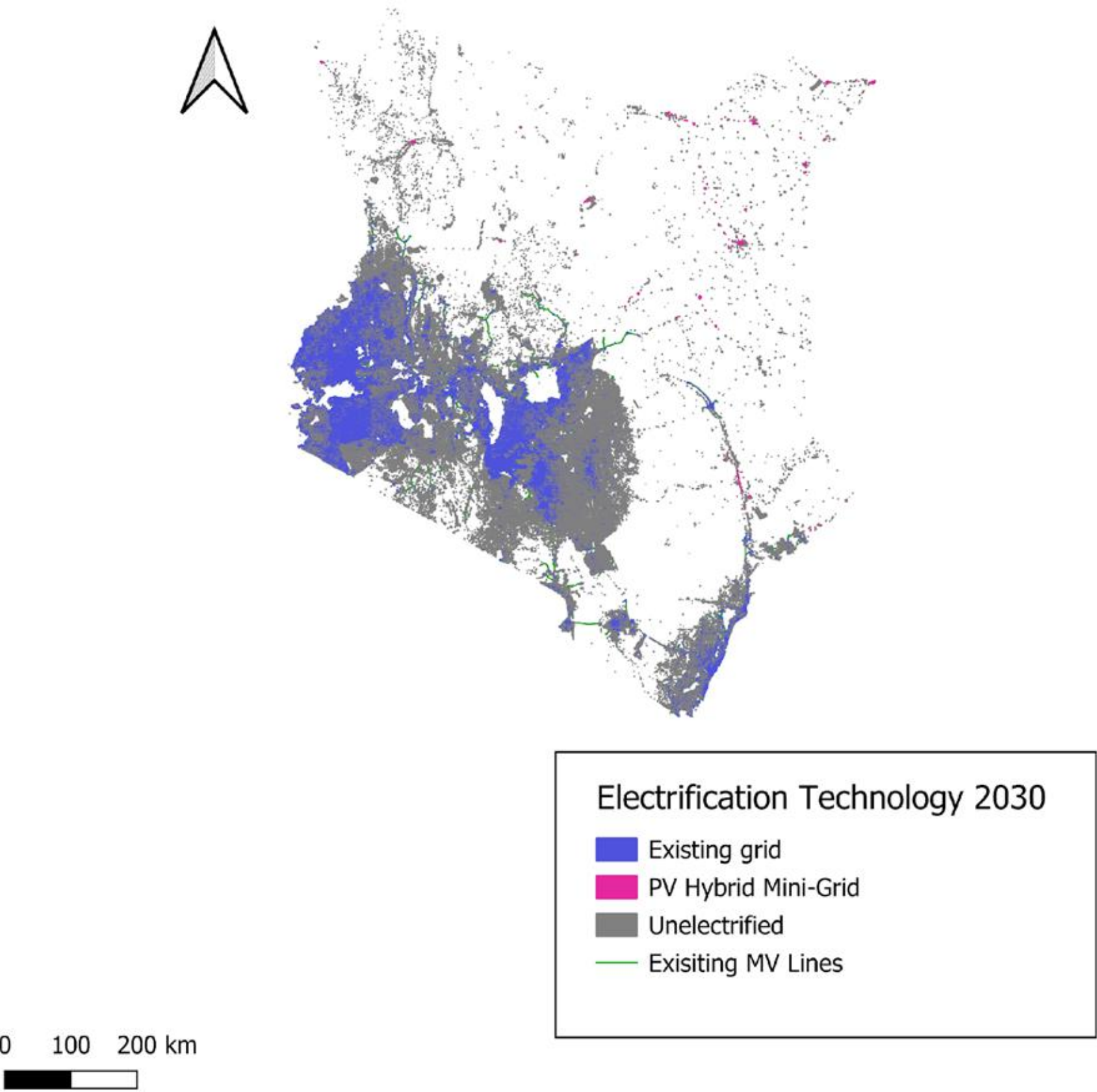
Kenya Electrification Pathways to 2030



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Energy Modelling Platform Africa (EMP-A)

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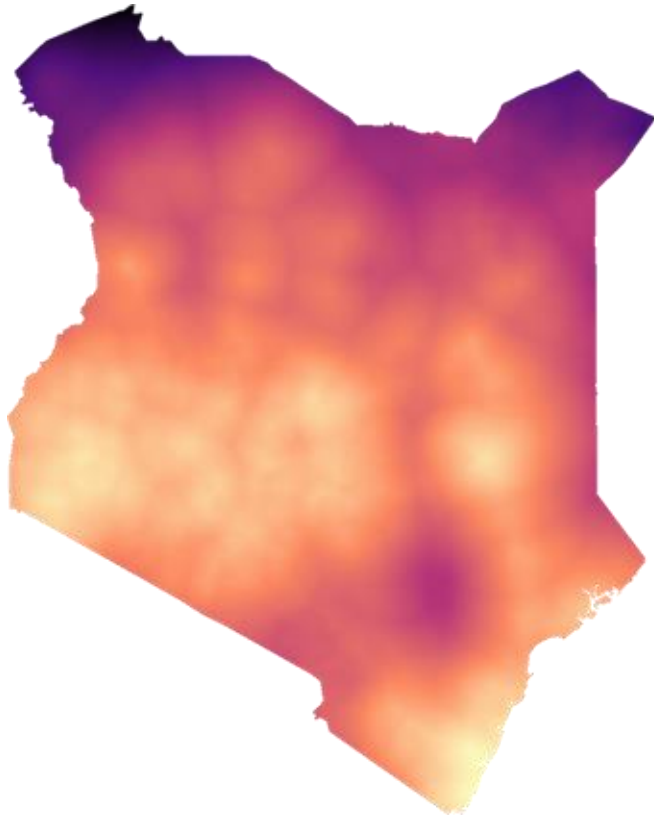
Category	Data
Population with Access	~75% national (Urban: ~90%, Rural: ~68%)
Population without Access	~10–11 million people (25%)
Main Electrification Technologies	Grid (~80%), Off-grid Decentralized Renewable Energy (~20%), 28.7% Geothermal, 26.5% Hydro, 13.4% Wind, 6.5% Solar, 18.7% Thermal
Generation capacity	Total Installed Capacity ~3,200 – 3,800 MW. Peak demand ~2,250 MW – 2,300 MW
Challenges	High rural extension costs, affordability, grid losses, weak reliability
Targets and Strategic Plans	Mission 300: Remote areas with mini-grids & DRE;>50% new connections via off-grid by 2030



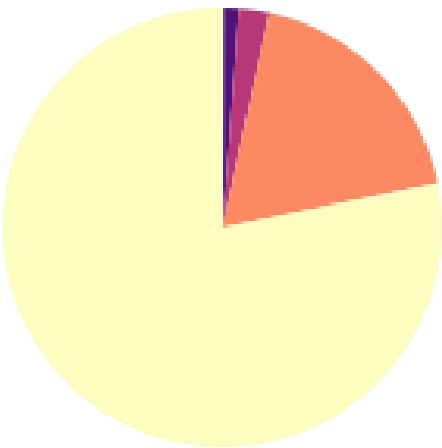
Objectives

Key Question	Kenya National Energy Compact Objective (Mission 300)
How do we reach universal access by 2030?	Connect ~5.1M households (grid + off-grid)
Where should off-grid scale fastest?	Prioritize remote, uneconomic grid areas
Where should investments be focused?	Use geospatial planning for prioritization
How do we ensure reliable access?	Strengthen transmission & distribution
What funding is required?	~USD 19B investment by 2030

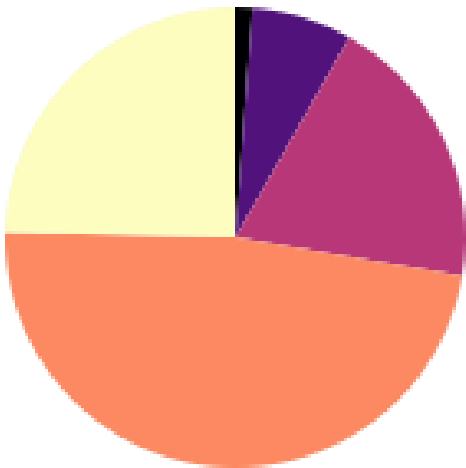
Kenya Energy Access Potential



Population Share



Area Share



Population in the analysis area: ~56.4 million people.

Population living within 1 km of an electricity line: ~61.3 million people

Source: ENERGY ACCESS EXPLORER

Scenarios

Using the OnSSET Model the following scenarios were investigated:

Scenario Label	Scenario Description	Key Assumptions (Scenario-Specific)
Grid-First Scenario	Grid expansion is the primary pathway to universal access by 2030.	• Grid-dominant rollout across most areas • No cap on annual grid connections • Grid prioritized within 2 km of MV lines if cost < USD 1,000/household • Off-grid solutions used only where grid is not feasible • Demand Tier Targets: Tier 4 (Urban), Tier 3 (Large Rural) and Tier 2 (Rural)
Least-Cost Scenario	Universal access achieved using the most affordable mix of technologies.	• Technology-neutral, least-cost selection • Grid expansion limited to cost-effective locations (up to 150 km) • Annual grid connections capped at 600,000 households • Mini-grids and solar used in remote or sparsely populated areas. • Demand Tier Targets: Tier 4 (Urban), Tier 3 (Large Rural) and Tier 2 (Rural)

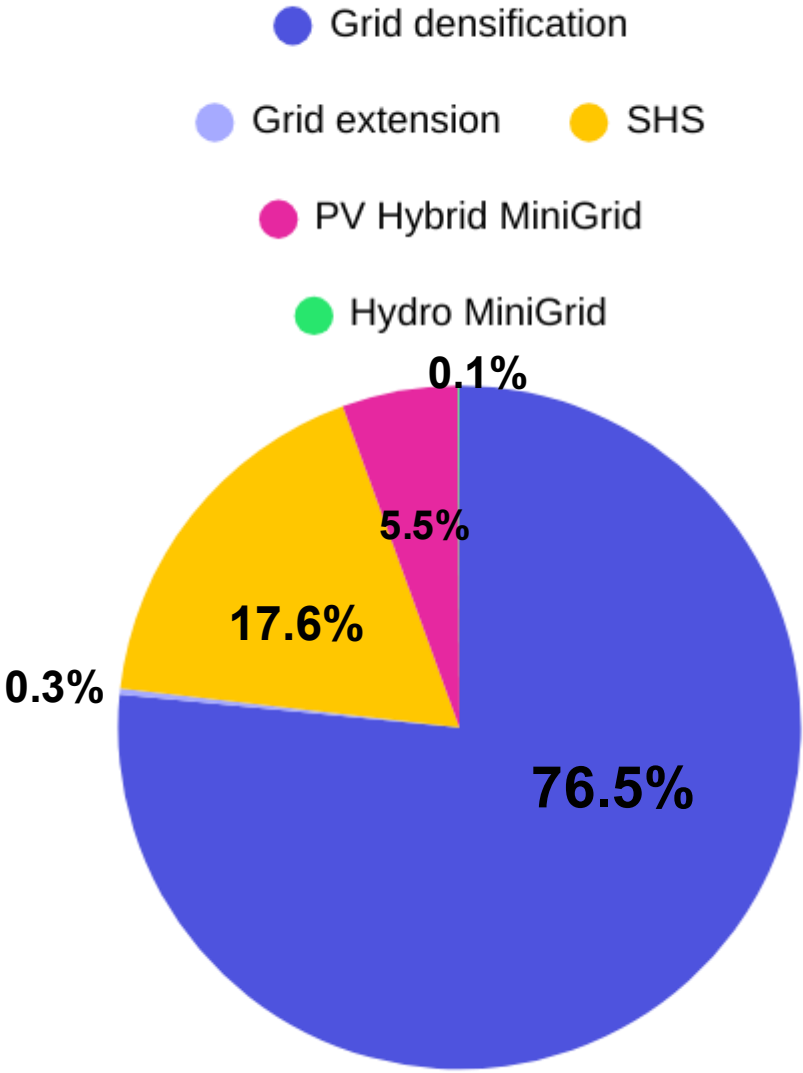
Option 1: Least-Cost Scenario

- Total investment cost: **USD 4.07 billion**
- Total installed capacity added: **~1,414 MW**
- Total emissions (2030): **~2,677 tCO₂**

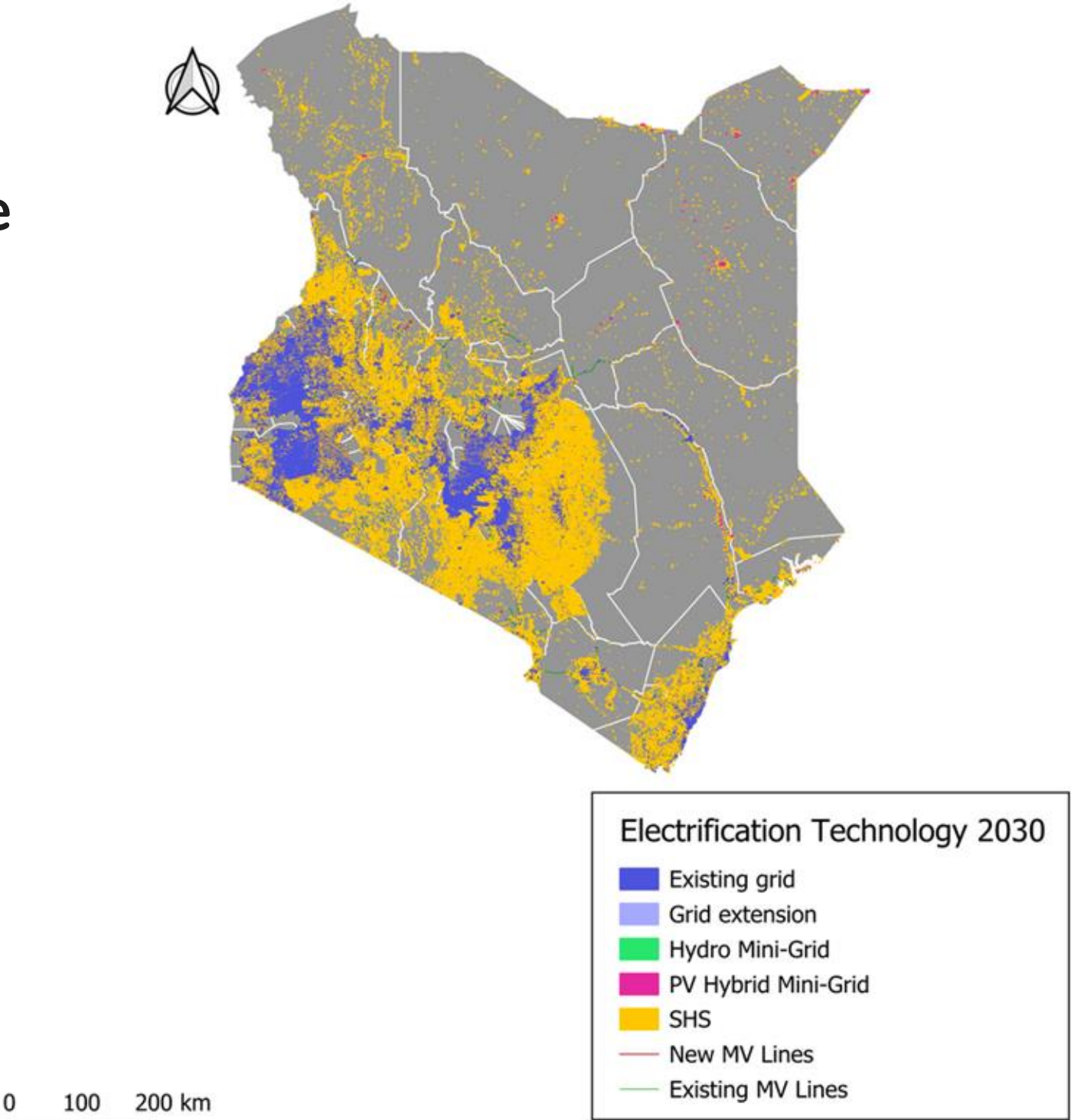
Grid vs Off-Grid Outcomes

- Total population connected to the grid: **44.40 million people**
- Total population served by off-grid solutions: **13.40 million people**

Population Estimation by technology 2030



Top Down High LCOE Electrification 2030



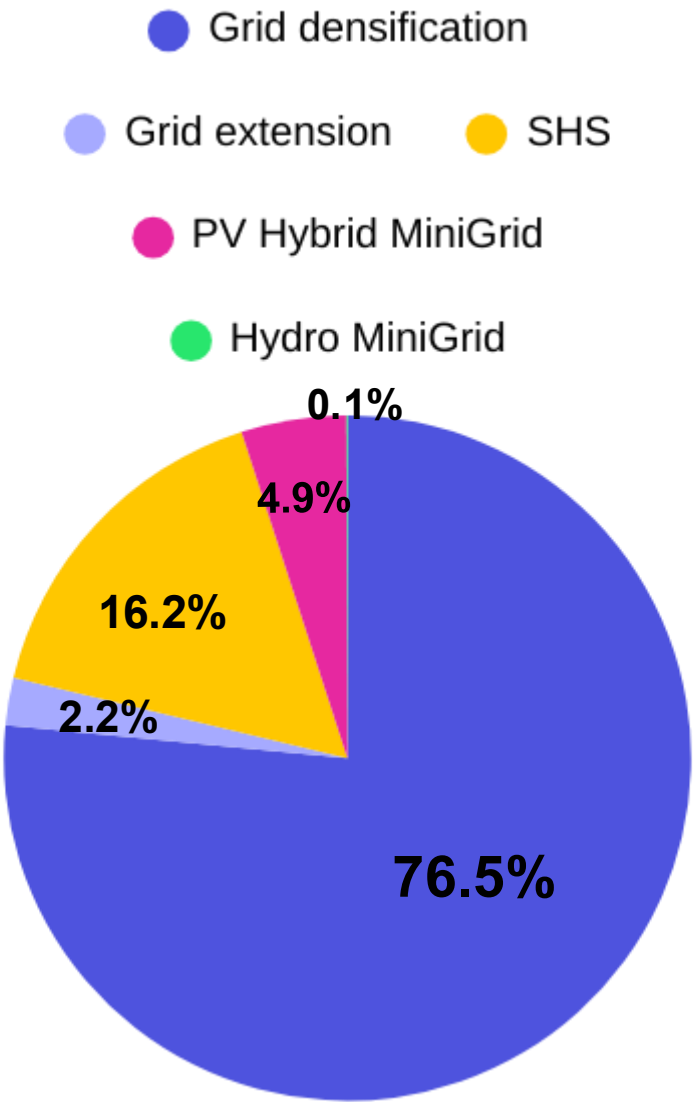
Option 2: Grid-First Scenario

- Total investment cost: **USD 4.39 billion**
- Total installed capacity added: **~1,408 MW**
- Total emissions (2030): **~2,524 tCO₂**

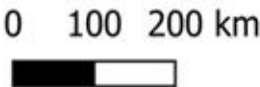
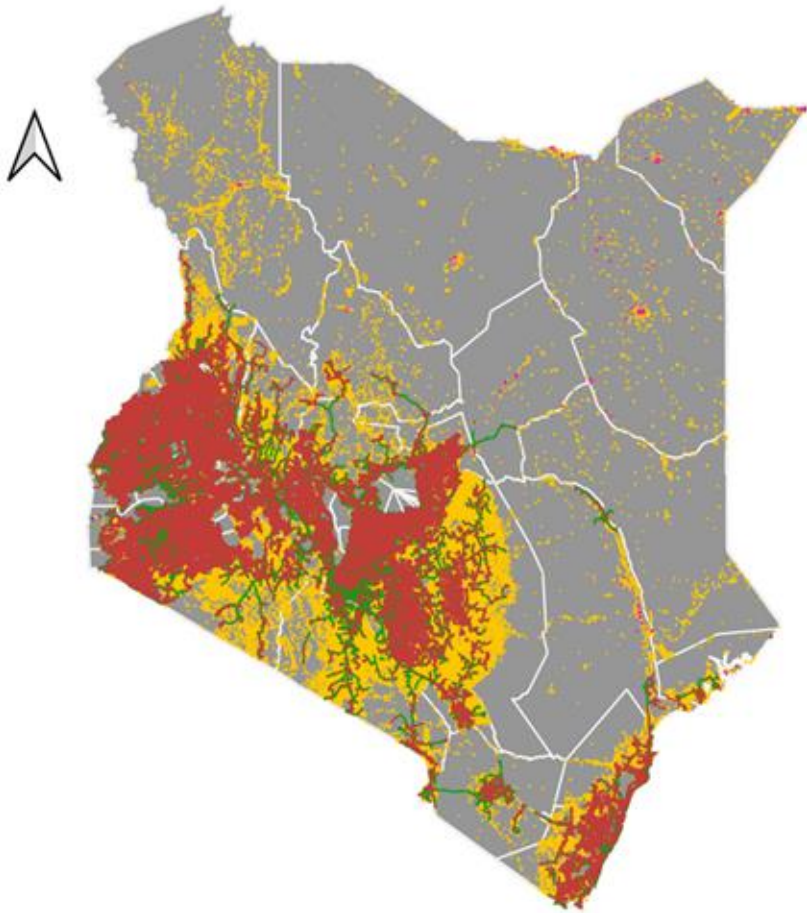
Grid vs Off-Grid Outcomes

- Total population connected to the grid: **45.53 million people**
- Total population served by off-grid solutions: **12.27 million people**

Population Estimation by technology 2030



Top Down High Forced Grid Electrification 2030



Policy Recommendation – Comparing Scenarios

Criteria	Grid-First High Demand Scenario	Least-Cost High Demand Scenario
Total Investment (Million USD)	USD 4.39B (~23.1% of the USD 19B)	USD 4.07B (saves ~USD 0.32B; ~21.3% of Kenya’s USD 19B)
Total Capacity (MW)	~1,408 MW	~1,414 MW (~6 MW more capacity)
Emissions (tCO ₂)	~2,524 tCO ₂	~2,677 tCO ₂
New Connections	~6.14 million	~6.14 million
Infrastructure Focus	Optimized mix: grid + mini-grids + standalone solar	Grid densification dominates; limited role for mini-grids and standalone solar
Implementation & Alignment	Public–private delivery; slightly higher upfront cost but lower emissions and system LCOE	Public-led, grid-centric; lower upfront spend but higher long-term risk and LCOE → Mission 300 aligned

Future Work

- **Align grid expansion with planned network upgrades** - Integrate transmission and distribution capacity constraints so model results reflect realistic grid expansion priorities for the Ministry of Energy.
- **Link modelling outputs to financing and implementation** - Use OnSSET results to support investment planning and financing discussions under Mission 300.

Thank You for Listening!